

Lujain Al-Momani

Dubai, UAE | lujain.momani@hotmail.com | +971 50 121 6373 | LinkedIn | Portfolio | Golden Visa Holder

Education

University of Wollongong in Dubai, Bachelor of Engineering in Telecommunication & IoT Dubai, UAE
Graduation: 2026 | GPA: 3.7/4

Experience

Research Intern – University of Wollongong (UOWD), Dubai, UAE Sept 2024 – Present

- Developed a lightweight machine-learning intrusion detection framework achieving 97% accuracy in constrained environments.
- Conducted analytical evaluation of alternative approaches and translated findings into practical system improvements with published outputs.

Clubs Director – Student Government, Rochester Institute of Technology (RIT), April 2023 – September 2023
Dubai

- Managed university clubs, led major events, and drove initiatives focused on student welfare and campus improvements.
- Built strong collaboration with students, faculty, and staff to successfully execute multiple high-impact projects.
- Nominated for Best Leadership Award and Best Club Award in recognition of outstanding contributions.

Business Intelligence Intern – Jordan Payments & Clearance Company July 2023 – August 2023
(JOPACC), Amman, Jordan

- Produced executive presentations and process manuals supporting adoption of new payments system.
- Built foundational skills in Power BI and ETL processes while enhancing internal reporting capabilities.

Publications

- Author of 5 research publications in AI, cybersecurity, and network optimization (ICCE 2025, FICTA 2025, ISDIA 2025, IEEEAccess; additional papers under review)

Projects

- **AstraIPS: AI-Assisted MQTT Intrusion Prevention System (Thesis)** — Built a fog-native, stateful IPS combining Snort 3, custom MQTT-aware Lua preprocessor, and BiLSTM model. Achieved 98% accuracy (AUC 0.9911) with <40ms latency on edge hardware. Winner, 1st Place – Research Category, UOWD. [GitHub](#) | [AstraIPS](#) — Fog-Native IoT Intrusion Prevention
- **Batterly: Smart Pancake Printer** — Led development of AI-assisted embedded pancake printer with computer vision and automated calibration.
- **Autonomous Waste Collection Robot (ERC 2025)** — Developed autonomous robotic system integrating perception and navigation for real-time decision making.
- **Hardware-Accelerated Intrusion Detection System for VANETs** — Prototyped real-time embedded security solution with 99.72% detection accuracy.

Skills & Certifications

Skills: Structured Problem Solving, Data Analysis & Visualization, Agile Project Management, Stakeholder Collaboration, Process Optimization, Risk Assessment, Strategic Thinking, Team Leadership

Tools: Python, MATLAB, Java, C, Power BI, ROS, Git, VMware, Cisco IOS, Machine Learning, Network Architecture & Security

Certifications: PwC Elevate Me (2025), McKinsey Forward Program (2024), Agile Software Development (2023)

Languages: Arabic (Native), English (Fluent)

Affiliations: IEEE, Engineers Australia

Competitions: HackMyRobot (NYUAD 2025), ERC (AUS 2025), Nexus 2.0 (Amity 2025), YouthxGE23